Growth and climate change
Stochastic IAMs in Dynare

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Materials: github.com/Vermandel/Macro-Modelling-Course-Nancy

Introduction

In what follows:

- I Introduce the basic conclusions that we can take from Nordhaus1992) from deterministic models:
  - Example of a deterministic/stochastic baby DICE
- Then I provide a short run perspective from Sahuc et al.2025) in which stabilization policies are also affected by mitigation policy in the short term
  - Example of a New Keynesian model

Why macroeconomists should care

- Climate change reshapes macroeconomic dynamics: productivity, inflation, employment.
- Monetary and fiscal policies are impacted through:
  - *Climateflation:* Damages reduce productivity, raise prices.
  - *Greenflation:* Carbon taxes raise costs during the transition.
- Macroeconomic tools must adapt: from stable-steady-state thinking to structural transitions.
- Need for integrated models that capture both physical and nominal frictions.

What is an Integrated Assessment Model (IAM)?

- IAMs link **economic activity** and **climate dynamics** to assess the costs and benefits of climate policies.
- The **IPCC** relies on a large pool of IAMs to produce long-term climate scenarios.
- These models are **highly heterogeneous**: many do not include optimization, forward-looking behavior, microfoundations, or expectations.
- The traditional benchmark is **DICE** (Nordhaus, 1992):
  - Embeds climate science into a **Ramsey-type growth model**.
  - Delivers optimal carbon pricing paths based on welfare maximization.

Why Climate Economics Needs a Short-Term View

- Policy institutions - central banks, treasuries, regulators - increasingly face the need to assess:
  - **Climate shocks** (e.g. disasters, weather volatility),
  - **Transition dynamics** (e.g. carbon tax, green investment),
  - **Stability risks** (price, financial, macroeconomic).
- This calls for models that are not only long-term and normative, but also **short-term and dynamic**.
- Yet, our standard macro tools are not designed for this:
  - DSGE models often focus on stationary dynamics near a steady state,
  - IAMs neglect shocks, expectations, and nominal frictions.
- This creates a **methodological gap** when addressing climate change as both a long-term trend and a short-term source of instability.

Outline

Plan

Introduction

- DICE (Dynamic Integrated Climate-Economy model) was developed by Nordhaus1992) and went through several (minor) updates up to the lastest 2023 snapshot;
- Simple but powerful model to provide fast estimates of the social cost of carbon and transition pathways for the world economy;
- It has become an intuitive benchmark.
- Back in the 90s, Bill Nordhaus's work on climate has never been plublished in top 5 journals (always rejected) but got published in Sciences, PNAS, etc.

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- The framework is old-fashion:
  - Solow growth model with climate externality;
  - Centralized equilibrium based on Ramsey allocations;
  - Purely deterministic setup;
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Carbon cycle

- Atmospheric loading of CO_2 is a system of differential equations:
$$M_t = \Phi_M M_{t-1} + \Xi_M E_t, \tag{1}$$
where $M_t = [M_t^{AT}, M_t^{UP}, M_t^{LO}]'$ comprises 3 layers of carbon: atmospheric, upper and lower ocean respectively.

- Matrices are determined by:
$$\Phi_M = \begin{bmatrix} 1-\phi_{12} & \phi_{21} & 0 \\ \phi_{12} & 1-\phi_{21}-\phi_{23} & \phi_{32} \\ 0 & \phi_{23} & 1-\phi_{32} \end{bmatrix} \text{ and } \Xi_M = \begin{bmatrix} \xi_M \\ 0 \\ 0 \end{bmatrix}$$

- Carbon emissions go into the atmosphere, before being captured by oceans through a long lasting process.

Temperatures

- Temperature anomalies $T_t = [T_t^{AT}, T_t^{OC}]$ of atmosphere and ocean respectively:

$$T_t = \Phi_T T_{t-1} + \Xi_T F_t, \tag{2}$$

- Matrices are given by:

$$\Phi_T = \begin{bmatrix} \varphi 11 & \varphi 12 \\ \varphi 21 & \varphi 22 \end{bmatrix} \text{ and } \Xi_T = \begin{bmatrix} \xi_T \\ 0 \end{bmatrix}$$

- Cooling effects of oceans through φ_{12}, ξ_T is the heating sensitivity to radiative forcing F_t.

Radiative forcing

- Greenhouse effect via radiative forcing (W/m^2):

$$F_t = \eta \ \log\left(M_t^{AT} / M_{1750}\right) / \log(2) + F_t^{EX} \tag{3}$$

- Here, η denote the forcing (Wm-2) of equilibrium CO2 doubling ($M_t^{AT} = 2M_{1750}$).
- F_t^{EX} is an exogenous process tracking the contribution of other sources of GHG (i.e. methane) on F_t

Input structure

- World population (billion):

$$L_t = L_{t-1} \left(L_T / L_{t-1} \right)^{\ell_g}, \tag{4}$$

$\ell_g \in [0,1]$ convergence rate, $L_T \in [0, +\infty)$ long term population.

- Law of motion of capital:

$$K_t = (1 - \delta_K) K_{t-1} + I_t, \tag{5}$$

$\delta_K \in [0,1]$ is the depreciation rate.

Production of goods and emissions

- Production is Cobb-Douglas:

$$Y_t = A_t K_{t-1}^{\gamma} L_t^{1-\gamma} \tag{6}$$

where $\gamma \in [0, 1]$ is capital intensity, exogenous TFP A_t, physical capital K_{t-1}.

- Emissions (GtCO2):

$$E_t = \sigma_t (1 - \mu_t) Y_t + E_t^{\text{land}} \tag{7}$$

where σ_t decoupling rate, μ_t is abatement share, E_t^{land} is exogenous process of carbon emission from change in land use (i.e. deforestation).

Equilibrium in goods and climate

- Resource constraint:

$$Y_t \left(1 - \theta_{1,t}\mu_t^{\theta_2}\right) \Omega\left(T_t\right) = C_t + I_t \tag{8}$$

LHS is output loss from abatement and damage, RHS is demand side.

- Damages are given by:

$$\Omega\left(T_t\right) = 1/(1 + aT_t^2) \tag{9}$$

This expression is subject to deep uncertainty, and should be taken as a normative measure of the social loss for the society of rising temperatures. Tipping points could be introduced [Weitzman2012].

Exogenous trends

Total factor productivity:

$$\Delta \log A_t = g_A - \delta_a \log \left(A_t / A_0 \right)$$

Decoupling rate of carbon emission to GDP:

$$\Delta \log \sigma_t = g_\sigma - \delta_\sigma \log \left(\sigma_t / \sigma_0 \right)$$

Cost of abating carbon emissions:

$$\theta_{1,t} = \left(1 - \delta_\theta \right) \theta_{1,t-1} \sigma_t / \sigma_{t-1}$$

Non-CO2 forcing law of motion:

$$F_t^{EX} = \min \left(F_{t-1}^{EX} + \Delta_F , F_{CAP} \right)$$

Land-use law of motion:

$$E_t^{\text{land}} = \left(1 - \delta_e \right) E_{t-1}^{\text{land}}$$

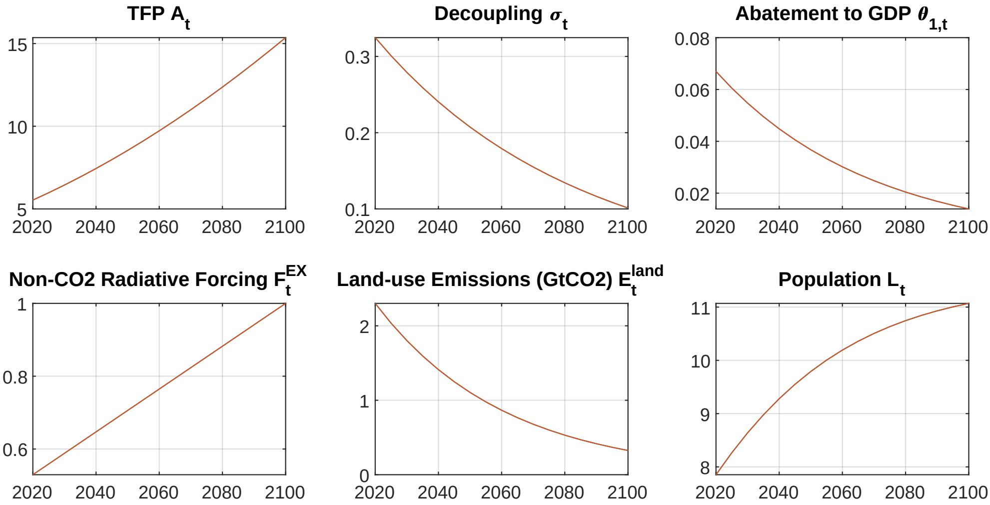


Figure. Exogenous variables

Introducing a saving rate

- How should we determine optimally C and I?

- Could use decentralized decision problem, or use old-fashion centralized equation following Ramsey allocation;

- Introduce optimal saving rate S, consumption and investment becomes:

$$C_t = (1 - S_t) \times Y_t \left(1 - \theta_{1,t}\mu_t^{\theta_2}\right) \Omega\left(T_t\right) \tag{10}$$

$$I_t = S_t \times Y_t \left(1 - \theta_{1,t}\mu_t^{\theta_2}\right) \Omega\left(T_t\right) \tag{11}$$

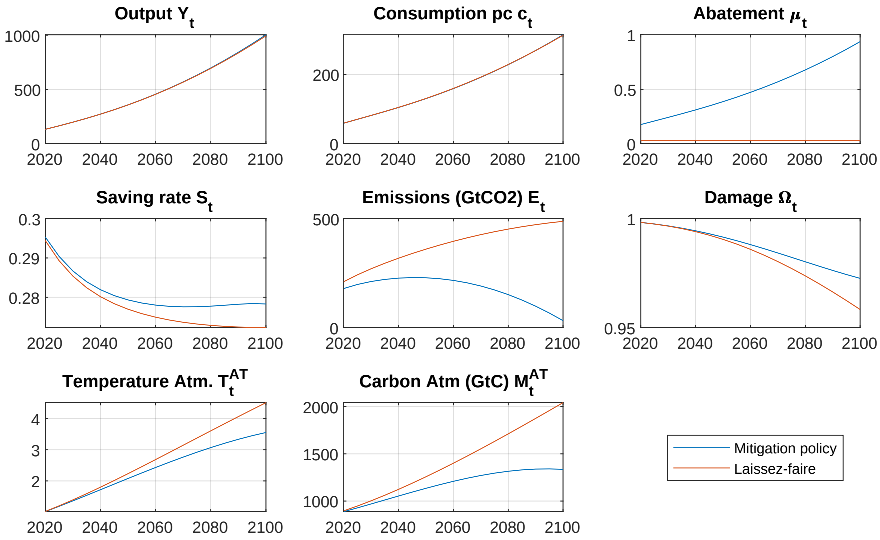
$\rightarrow$implicitely assuming that income spent between C_t and I_t.

Introducing a saving rate

- The planner maximizes the social welfare:

$$\max_{\{S_t,\mu_t\}} \sum_{t=0}^{\infty} \beta^t L_t \left(c_t^{1-\alpha} - 1\right)(1-\alpha)^{-1} \tag{12}$$

where $c_t = C_t/L_t$ is the per capita consumption, utility function concave.


Output Y_t
1000
500
0
Consumption pc c_t
200
0
Abatement μ_t
1
0.5
0
Saving rate S_t
0.3
0.29
0.28
Emissions (GtCO2) E_t
500
0
Damage Ω_t
1
0.95
Temperature Atm. T_t^{AT}
4
3
2
Carbon Atm (GtC) M_t^{AT}
2000
1500
1000
2020 2040 2060 2080 2100
Mitigation policy
Laissez-faire

Some striking but controversial results from DICE:

- "Optimal" to warm the planet up to 3.5°C;
- Mitigation policy saves about 2% of climate damage;
- Net zero emission optimal by 2100, 50 years later than in Paris-Agreement;
- Main critic from DICE's policy output: those results have scientifically grounded climate inaction in policy circles...

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$\rightarrow$Let's work on the exercise 1 & 2 of PC2
and baby_rbc1.mlx, baby_rbc2.mlx, baby_rbc3.mlx

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- Climate change will shake the macroeconomic landscape in the next decades and the central bank will have to face 2 phenomena [Schnabel2022]:
  - On the one hand, a warming planet causes damages that will make resources scarcer & prices higher → climateflation.
  - On the other hand, the fight against climate change through increasing carbon taxes will increase production costs → greenflation.
- How should the central bank conduct monetary policy in this new landscape?
- Answering this question requires a new class of IAM with New Keynesian ingredients to capture inflation dynamics.
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  - extending the New Keynesian model with a **carbon accumulation constraint** and a **mitigation policy** from the Integrated Assessment Model (IAM) [Barrage and Nordhaus2023];
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- Standard view: stable propagation mechanism with fluctuations naturally decaying over time back to a steady state [Smets and Wouters2007].
- Climate problem: Cumulative emissions permanently alter the propagation mechanism $\rightarrow$ no steady state.
- Why? Cumulative carbon induces permanent shifts in productivity, structurally similar to an endogenous (de-)growth model.
- Climate damages $\Omega(M)$ in production ($Y = \Phi(M) \cdot L$) does not scale proportionally with economic growth, violating the homogeneity assumption required for Solow's detrending techniques.
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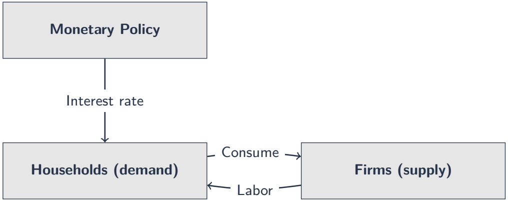

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- Population dynamics $l_t = l_{t-1}^{(1-\ell)} l_\infty^\ell$ (exogenous)

- Abatement technology (exogenous)
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- TFP $z_t = z_{t-1}(1 + g_{z,t})$ with $g_{z,t} = g_{z,t-1}(1 - \delta_z)$ (exogenous)

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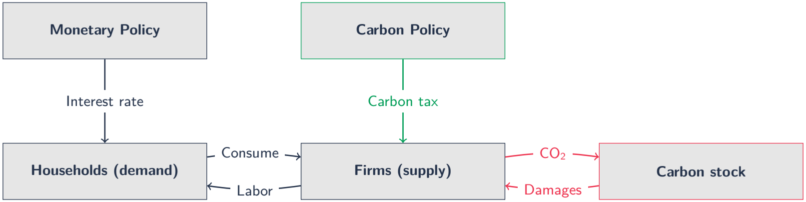

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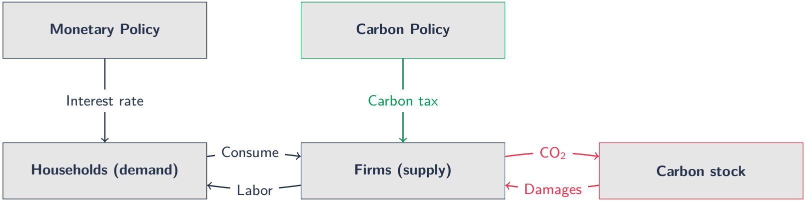

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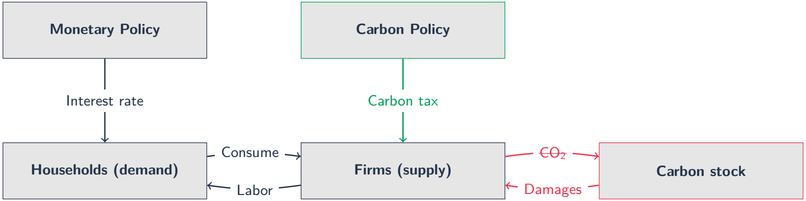

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The New Keynesian Climate Model

Carbon accumulation and its damages:

IS:
$$\left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega)\left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c}\right)$$
$$x_t = 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta(1 - \varepsilon_{p,t} mc_t)$$

PC:
$$\left(\pi_t - \pi_t^*\right) \pi_t = (1-\vartheta)\beta \mathbb{E}_t\, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta)$$
$$mc_t = \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1}\, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha}$$

MP:
$$r_t = r_{t-1}^{\rho} \left[r_r \pi_t^* \left(\pi_t/\pi_t^*\right)^{\phi_\pi} \left(\tilde{y}_t/\tilde{y}_t^n\right)^{\phi_y}\right]^{1-\rho} \varepsilon_{r,t}$$

CC:
$$\tilde{m}_t = \tilde{m}_{t-1} + \xi_m\, \sigma_t\, z_t\, l_t\, \tilde{y}_t\, \varepsilon_{e,t}$$

The New Keynesian Climate Model

Carbon accumulation and its damages:

$$
\begin{aligned}
\textbf{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} &= \beta \mathbb{E}_t \, \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right) \\
x_t &= 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t) \\
\textbf{PC:} \quad \left(\pi_t - \pi_t^*\right) \pi_t &= (1-\vartheta) \beta \mathbb{E}_t \, g_{z,t} \tilde{y}_{t+1} / \tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1-\zeta) \\
mc_t &= \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1} \, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha} \\
\textbf{MP:} \quad r_t &= r_{t-1}^{\rho} \left[r_r \pi_t^* \left(\pi_t / \pi_t^*\right)^{\phi_\pi} \left(\tilde{y}_t / \tilde{y}_t^n\right)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t} \\
\textbf{CC:} \quad \tilde{m}_t &= \tilde{m}_{t-1} + \xi_m \, \sigma_t \, z_t \, l_t \, \tilde{y}_t \, \varepsilon_{e,t}
\end{aligned}
$$

Anthropogenic carbon stock

The New Keynesian Climate Model

Carbon accumulation and its damages:

IS:
$$\left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t)$$

PC:
$$\left(\pi_t - \pi_t^*\right) \pi_t = (1-\vartheta)\beta \mathbb{E}_t \, g_{z,t} \tilde{y}_{t+1} / \tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1-\zeta)$$

$$mc_t = \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1} \, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha}$$

MP:
$$r_t = \ldots \left(\ldots\right)^{\phi_\pi} \left(\tilde{y}_t / \tilde{y}_t^n\right)^{\phi_y} \Big]^{1-\rho} \varepsilon_{r,t}$$

Deterministic Decoupling trend

CC:
$$\tilde{m}_t = \tilde{m}_{t-1} + \xi_m \, \sigma_t \, z_t \, l_t \, \tilde{y}_t \, \varepsilon_{e,t}$$

Anthropogenic carbon stock

The New Keynesian Climate Model

Carbon accumulation and its damages:

$$\textbf{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega)\left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta(1 - \varepsilon_{p,t} mc_t)$$

$$\textbf{PC:} \quad \left(\pi_t - \pi_t^*\right)\pi_t = (1-\vartheta)\beta \mathbb{E}_t \, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right)\pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta)$$

$$mc_t = \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1} \, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha}$$

Deterministic Decoupling trend

$$\textbf{MP:} \quad r_t = \ldots {}^{\phi_\pi} \left(\tilde{y}_t / \tilde{y}_t^n\right)^{\phi_y} \Big]^{1-\rho} \varepsilon_{r,t}$$

$$\textbf{CC:} \quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m \, \sigma_t \, z_t \, l_t \, \tilde{y}_t \, \varepsilon_{e,t}$$

Anthropogenic carbon stock

Deterministic

The New Keynesian Climate Model

Carbon accumulation and its damages:

IS:
$$\left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t)$$

PC:
$$\left(\pi_t - \pi_t^*\right)\pi_t = (1-\vartheta)\beta \mathbb{E}_t\, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right)\pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta)$$

$$mc_t = \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1}\, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha}$$

MP: r_t

Deterministic Decoupling trend

Deterministic population trend

CC:
$$\tilde{m}_t = \tilde{m}_{t-1} + \xi_m\, \sigma_t\, z_t\, l_t\, \tilde{y}_t\, \varepsilon_{e,t}$$

Anthropogenic carbon stock

Deterministic

The New Keynesian Climate Model

Carbon accumulation and its damages:

$$
\begin{aligned}
\textbf{IS:}\quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} &= \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega)\left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c}\right) \\
x_t &= 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta(1 - \varepsilon_{p,t} mc_t)
\end{aligned}
$$

$$
\begin{aligned}
\textbf{PC:}\quad \left(\pi_t - \pi_t^*\right)\pi_t &= (1-\vartheta)\beta \mathbb{E}_t\, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right)\pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta) \\
mc_t &= \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1}\, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha}
\end{aligned}
$$

MP: r_t … ϕ_π …

Deterministic Decoupling trend

Deterministic population trend

Emission AR(1) shock

$$
\textbf{CC:}\quad \tilde{m}_t = \tilde{m}_{t-1} + \xi_m\, \sigma_t\, z_t\, l_t\, \tilde{y}_t\, \varepsilon_{e,t}
$$

Anthropogenic carbon stock

Deterministic

The New Keynesian Climate Model

Carbon accumulation and its damages:

$$
\begin{aligned}
\textbf{IS:}\quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} &= \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega)\left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c}\right) \\
x_t &= 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta\left(1 - \varepsilon_{p,t} mc_t\right) \\
\textbf{PC:}\quad \left(\pi_t - \pi_t^*\right)\pi_t &= (1-\vartheta)\beta \mathbb{E}_t\, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right)\pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta) \\
mc_t &= \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1}\, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha} \longleftarrow \text{Climate damages} \\
\textbf{MP:}\quad r_t &= r_{t-1}^{\rho} \left[r_r \left(\pi_t^*/\pi\right)\left(\pi_t/\pi_t^*\right)^{\phi_\pi} \left(\tilde{y}_t/\tilde{y}_t^n\right)^{\phi_y}\right]^{1-\rho} \varepsilon_{r,t} \\
\textbf{CC:}\quad \tilde{m}_t &= \tilde{m}_{t-1} + \xi_m\, \sigma_t\, z_t\, l_t\, \tilde{y}_t\, \varepsilon_{e,t}
\end{aligned}
$$

The New Keynesian Climate Model

Mitigation policies as function of exogenous carbon tax $\tilde{\tau}_t$:

Mitigation expenditures

$$
\begin{aligned}
\textbf{IS:} \quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} &= \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right) \\
x_t &= 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t) - \theta_{1,t} \tilde{\tau}_t^{\theta_2/(\theta_2-1)} \\
\textbf{PC:} \quad \left(\pi_t - \pi_t^*\right) \pi_t &= (1-\vartheta) \beta \mathbb{E}_t \, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1-\zeta) \\
mc_t &= \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1} \, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha} + \theta_{1,t} \tilde{\tau}_t \left(\theta_2 + (1-\theta_2) \tilde{\tau}_t^{1/(\theta_2-1)}\right) \\
\textbf{MP:} \quad r_t &= r_{t-1}^{\rho} \left[r_r \left(\pi_t^*/\pi\right) \left(\pi_t/\pi_t^*\right)^{\phi_\pi} \left(\tilde{y}_t/\tilde{y}_t^n\right)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t} \\
\textbf{CC:} \quad \tilde{m}_t &= \tilde{m}_{t-1} + \xi_m \, \sigma_t \, z_t \, l_t \, \tilde{y}_t \, \varepsilon_{e,t} \left(1 - \tilde{\tau}_t^{1/(\theta_2-1)}\right)
\end{aligned}
$$

The New Keynesian Climate Model

Mitigation policies as function of exogenous carbon tax $\tilde{\tau}_t$:

Mitigation expenditures

Carbon tax costs

$$
\begin{aligned}
\textbf{IS:}\quad \left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} &= \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right) \\
x_t &= 1 - (1-\vartheta) 0.5 \kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta (1 - \varepsilon_{p,t} mc_t) - \theta_{1,t} \tilde{\tau}_t^{\theta_2/(\theta_2-1)} \\
\textbf{PC:}\quad \left(\pi_t - \pi_t^*\right) \pi_t &= (1-\vartheta) \beta \mathbb{E}_t \, g_{z,t} \tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right) \pi_{t+1} + \zeta \kappa^{-1} \varepsilon_{p,t} mc_t + \kappa^{-1} (1-\zeta) \\
mc_t &= \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1} \, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha} + \theta_{1,t} \tilde{\tau}_t \left(\theta_2 + (1-\theta_2) \tilde{\tau}_t^{1/(\theta_2-1)}\right) \\
\textbf{MP:}\quad r_t &= r_{t-1}^{\rho} \left[r_r \left(\pi_t^*/\pi\right) \left(\pi_t/\pi_t^*\right)^{\phi_\pi} \left(\tilde{y}_t/\tilde{y}_t^n\right)^{\phi_y} \right]^{1-\rho} \varepsilon_{r,t} \\
\textbf{CC:}\quad \tilde{m}_t &= \tilde{m}_{t-1} + \xi_m \, \sigma_t \, z_t \, l_t \, \tilde{y}_t \, \varepsilon_{e,t} \left(1 - \tilde{\tau}_t^{1/(\theta_2-1)}\right)
\end{aligned}
$$

The New Keynesian Climate Model

Mitigation policies as function of exogenous carbon tax $\tilde{\tau}_t$:

Mitigation expenditures

IS: $$\left(\frac{\tilde{y}_t x_t - \omega d}{1-\omega}\right)^{-\sigma_c} = \beta \mathbb{E}_t \frac{\varepsilon_{b,t+1}}{\varepsilon_{b,t}} \frac{r_t}{\pi_{t+1}} \left((1-\omega) \left(\frac{x_{t+1}\tilde{y}_{t+1} - \omega d}{1-\omega}\right)^{-\sigma_c} + \omega d^{-\sigma_c} \right)$$

$$x_t = 1 - (1-\vartheta) 0.5\kappa \left(\pi_t - \pi_t^*\right)^2 - \vartheta(1 - \varepsilon_{p,t} mc_t) - \theta_{1,t}\tilde{\tau}_t^{\theta_2/(\theta_2-1)}$$

Carbon tax costs

PC: $$\left(\pi_t - \pi_t^*\right)\pi_t = (1-\vartheta)\beta\mathbb{E}_t\, g_{z,t}\tilde{y}_{t+1}/\tilde{y}_t \left(\pi_{t+1} - \pi_{t+1}^*\right)\pi_{t+1} + \zeta\kappa^{-1}\varepsilon_{p,t} mc_t + \kappa^{-1}(1-\zeta)$$

$$mc_t = \psi \left(x_t \tilde{y}_t - \omega d\right)^{\sigma_c} \tilde{y}_t^{(1+\sigma_n)/\alpha - 1}\, \Phi\left(\tilde{m}_t\right)^{-(1+\sigma_n)/\alpha} + \theta_{1,t}\tilde{\tau}_t\left(\theta_2 + (1-\theta_2)\tilde{\tau}_t^{1/(\theta_2-1)}\right)$$

Abatement share

MP: $$r_t = r_{t-1}^{\rho}\left[r_r \left(\pi_t^*/\pi\right)\left(\pi_t/\pi_t^*\right)^{\phi_\pi}\left(\tilde{y}_t/\tilde{y}_t^n\right)^{\phi_y}\right]^{1-\rho}$$

CC: $$\tilde{m}_t = \tilde{m}_{t-1} + \xi_m\, \sigma_t\, z_t\, l_t\, \tilde{y}_t\, \varepsilon_{e,t}\left(1 - \tilde{\tau}_t^{1/(\theta_2-1)}\right)$$

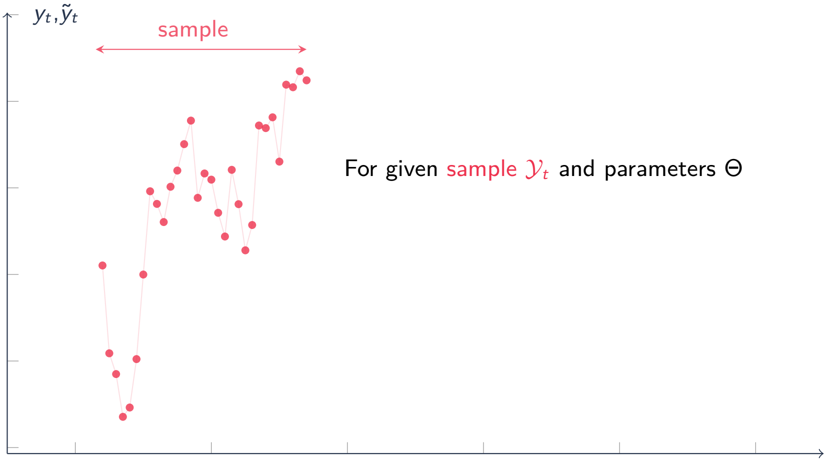
Estimation

- Estimation on world data from 1985Q1 to 2023Q3 (sources: World Bank, OECD and OurWorldInData).
- There are four observable variables:

$$\begin{bmatrix} \text{Real output growth rate} \\ \text{Inflation rate} \\ \text{Short-term interest rate} \\ CO_2 \text{ emissions growth rate} \end{bmatrix} = 100 \times \begin{bmatrix} \Delta \log(y_t) \\ \pi_t - 1 \\ r_t - 1 \\ \Delta \log(e_t) \end{bmatrix}$$

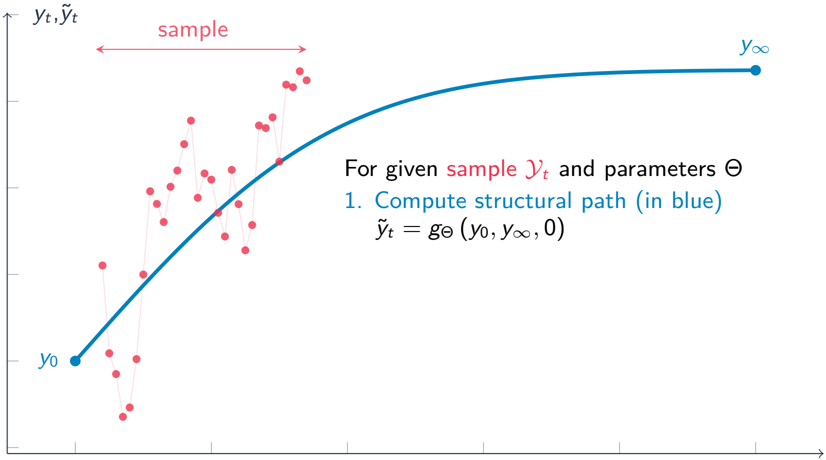
- Solution & filtering methods from Fair and Taylor1983): fully nonlinear, MIT shocks and no aggregate uncertainty.

Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

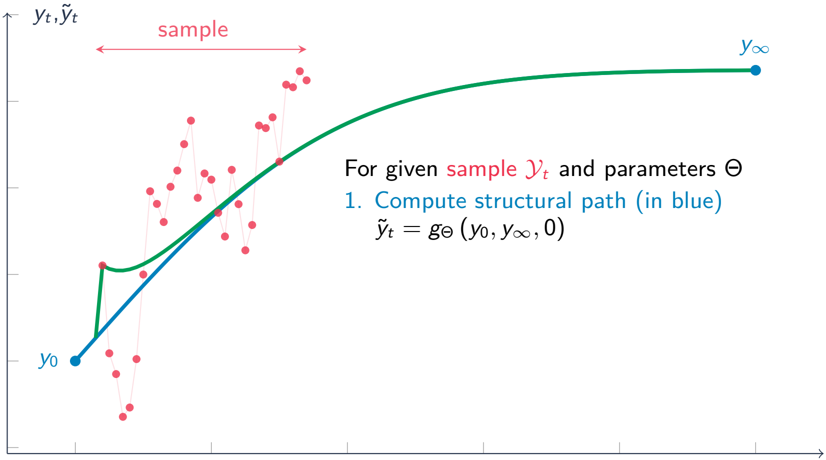
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta(y_0, y_\infty, 0)$

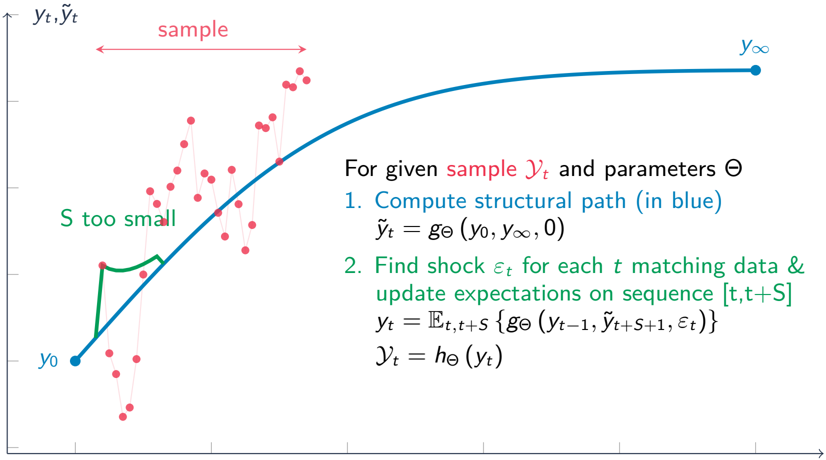
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta(y_0, y_\infty, 0)$

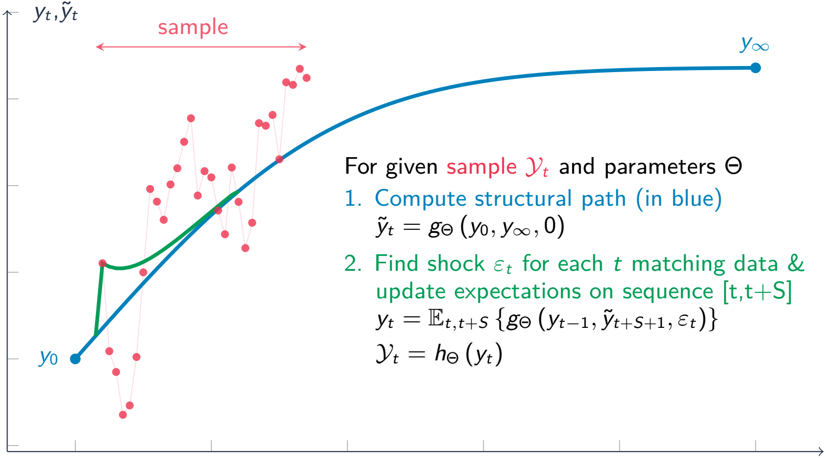
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta(y_0, y_\infty, 0)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\{g_\Theta(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t)\}$
   $\mathcal{Y}_t = h_\Theta(y_t)$

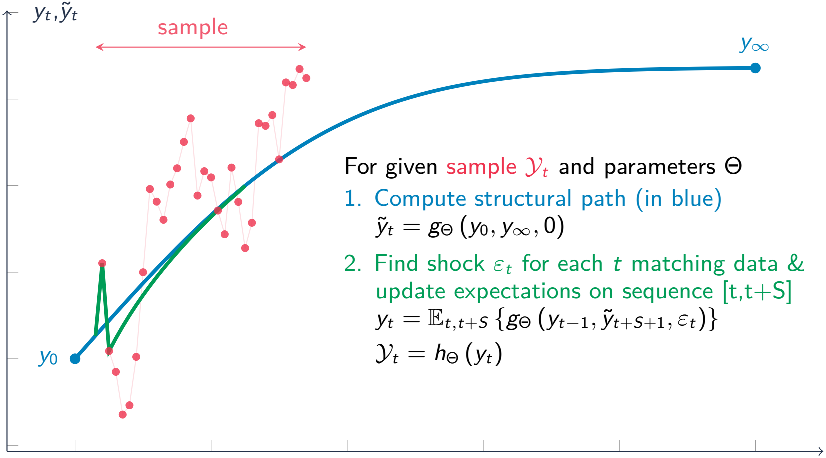
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta\left(y_0, y_\infty, 0\right)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\left\{g_\Theta\left(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t\right)\right\}$
   $\mathcal{Y}_t = h_\Theta\left(y_t\right)$

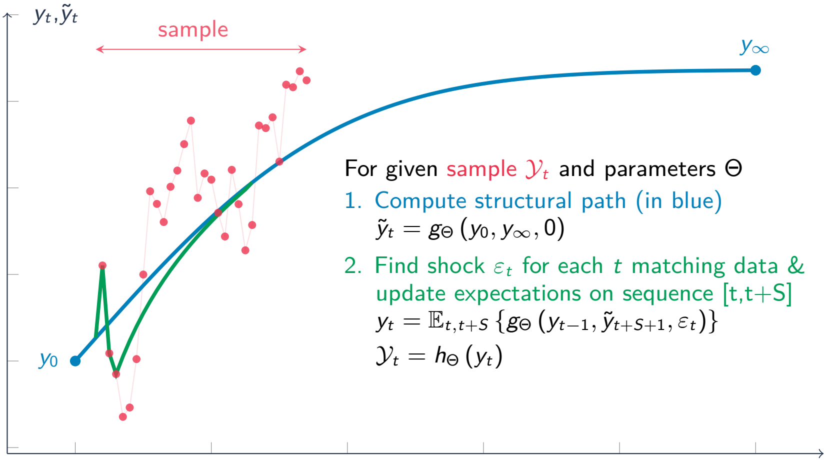
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta(y_0, y_\infty, 0)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\{g_\Theta(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t)\}$
   $\mathcal{Y}_t = h_\Theta(y_t)$

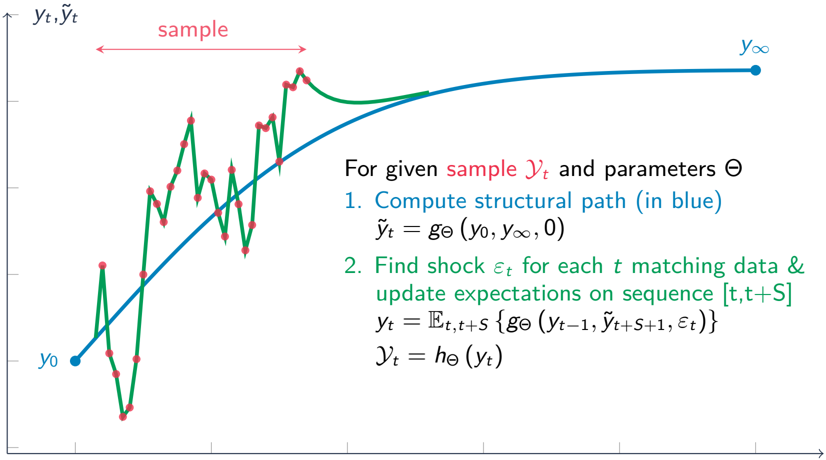
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta (y_0, y_\infty, 0)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S} \{ g_\Theta (y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t) \}$
   $\mathcal{Y}_t = h_\Theta (y_t)$

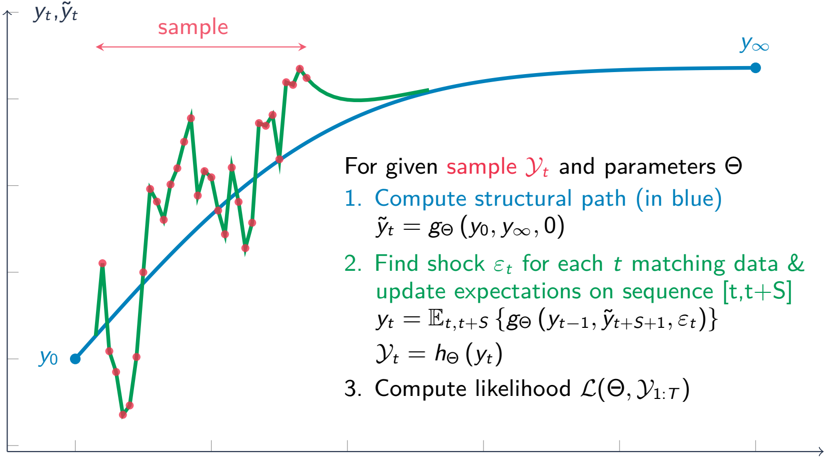
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta\left(y_0, y_\infty, 0\right)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\left\{g_\Theta\left(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t\right)\right\}$
   $\mathcal{Y}_t = h_\Theta\left(y_t\right)$

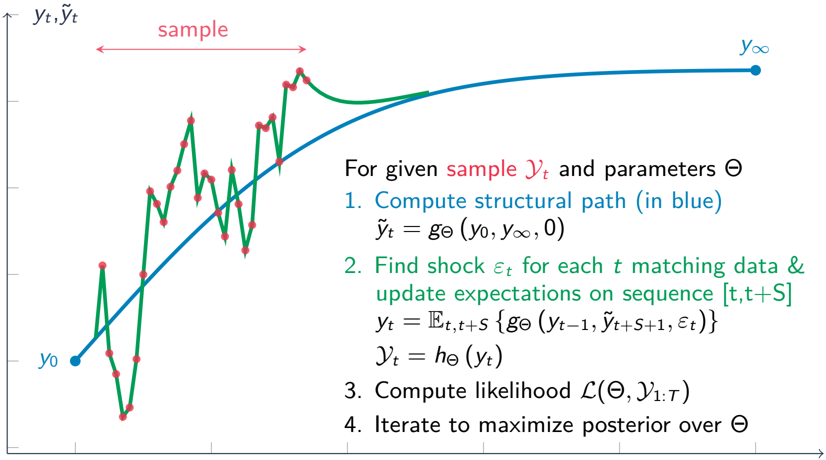
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta\left(y_0, y_\infty, 0\right)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\left\{g_\Theta\left(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t\right)\right\}$
   $\mathcal{Y}_t = h_\Theta\left(y_t\right)$
3. Compute likelihood $\mathcal{L}(\Theta, \mathcal{Y}_{1:T})$

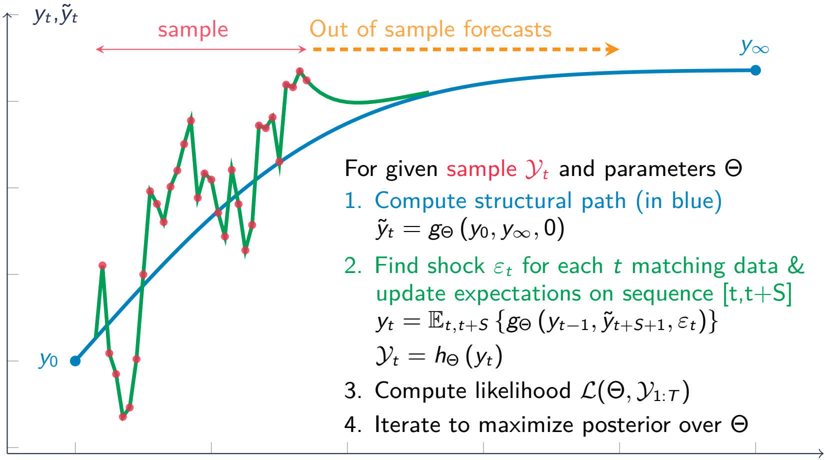
Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta \left(y_0, y_\infty, 0 \right)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S} \left\{ g_\Theta \left(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t \right) \right\}$
   $\mathcal{Y}_t = h_\Theta \left(y_t \right)$
3. Compute likelihood $\mathcal{L}(\Theta, \mathcal{Y}_{1:T})$
4. Iterate to maximize posterior over Θ

Estimation



For given sample $\mathcal{Y}_t$ and parameters Θ

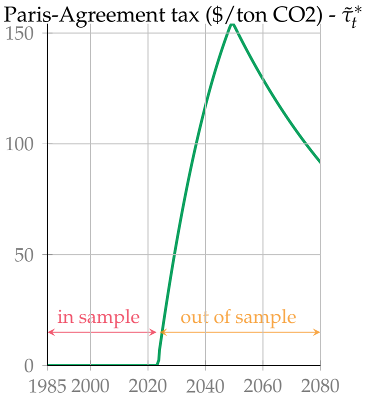
1. Compute structural path (in blue)
   $\tilde{y}_t = g_\Theta\left(y_0, y_\infty, 0\right)$
2. Find shock ε_t for each t matching data & update expectations on sequence [t,t+S]
   $y_t = \mathbb{E}_{t,t+S}\left\{g_\Theta\left(y_{t-1}, \tilde{y}_{t+S+1}, \varepsilon_t\right)\right\}$
   $\mathcal{Y}_t = h_\Theta\left(y_t\right)$
3. Compute likelihood $\mathcal{L}(\Theta, \mathcal{Y}_{1:T})$
4. Iterate to maximize posterior over Θ

Estimation

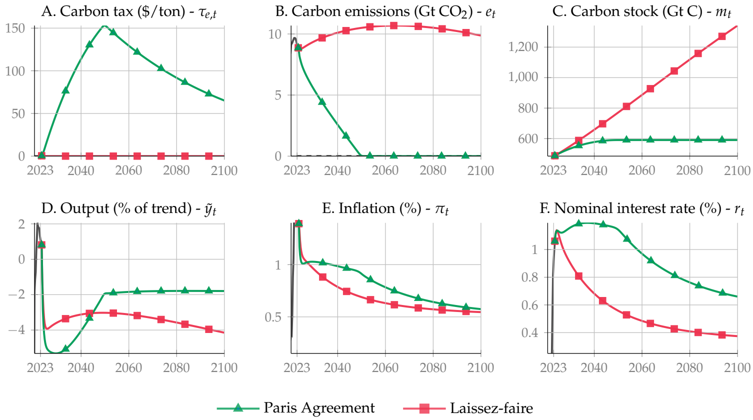
- Large uncertainty about future carbon tax: implications for estimation in particular at the end of the sample
- Let $\tilde{\tau}_t^*$ denote the Paris-Agreement tax, with linear decrease in carbon emissions up to 2050
- We let the data inform about future carbon mitigation policies:

$$\mathbb{E}_t\{\tilde{\tau}_t\} = \varphi\tilde{\tau}_t^*$$

where $\varphi \in [0,1]$ is the fraction of believers (or expected intensity) of Paris-Agreement policy.



Three Transitions



Dissecting the PC curve

- How big is climateflation versus greenflation?
- We split the marginal cost into three terms:
- Each force expressed in inflation-equivalent terms $\hat{\pi}_t^c$ and $\hat{\pi}_t^g$

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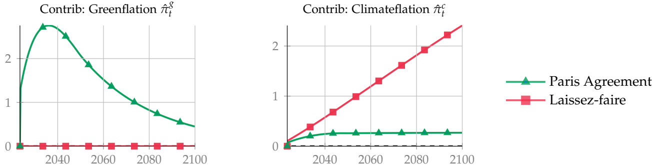
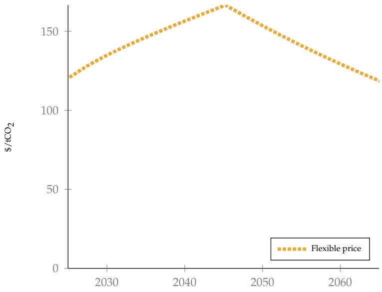


Why nominal rigidities matter for the SCC

What are the implications of nominal rigidities for the social cost of carbon (SCC)?



- Under flexible prices, the planner can adjust relative prices instantly
- Under sticky prices, costs cannot adjust immediately, so the economy adjusts through quantities
- Lower consumption c_t under nominal rigidities means the planner is less willing to use the tax instrument

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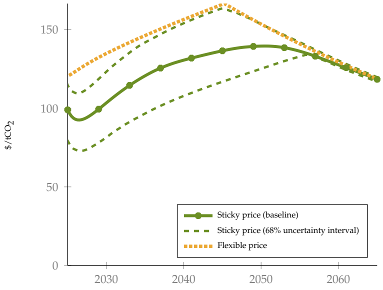

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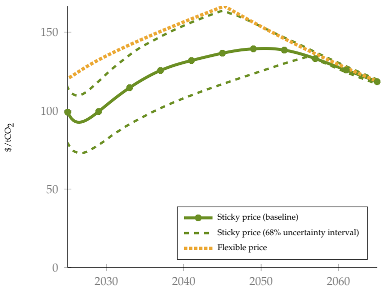

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Conclusion

- climateflation and greenflation are sizable, but their magnitude depends crucially on how real wages adjust.
- Greenflation is relatively robust to capital, tax redistribution, sticky wages, and optimal taxing.
- Nominal rigidities matter for the social cost of carbon (SCC): IAMs should reflect the nominal structure of the economies they are meant to guide.

Thank you for your attention

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